Fiscal Year:	FY 2011	**Task Last Updated:**	FY 11/09/2010
PI Name:	Lockley, Steven W Ph.D.		
Project Title:	Characteristics of Light Exposure Necessary for Development of Optimal Countermeasures to Facilitate Circadian Adaptation and Enhance Alertness and Cognitive Performance in Space		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02115-5804	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2007 NSBRI-RFA-07-01 Human Health in Space
Start Date:	10/01/2007	**End Date:**	09/30/2011
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ronda, Joseph (Brigham and Women's Hospital) Aeschbach, Daniel (Brigham and Women's Hospital) Gooley, Joshua (Brigham and Women's Hospital)		
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Task Description:	Space missions often expose crew members to unusual light-dark cycles that lead to misalignment between the circadian pacemaker and the sleep-wake schedule, resulting in disturbed sleep and impaired waking function. Sleep disturbances due to other reasons (e.g., noise, temperature, microgravity) further exacerbate cognitive deficits and these factors jointly increase the risk of fatigue-related accidents and injuries. An effective countermeasure is required to facilitate more rapid adaptation of the circadian system and directly enhance alertness and performance. Light exposure induces both of these effects and is an effective, safe, non-pharmacological countermeasure for circadian- and fatigue-related deficits in cognition. Recently, we and others have shown that short-wavelength (blue) light exposure at night is the most effective wavelength for phase-shifting the circadian pacemaker, suppressing pineal melatonin, enhancing subjective alertness, improving performance and inducing EEG-derived brain activation that indicates a more alert state. These effects, however, have only been examined during the night using relatively bright light and recent evidence suggests that the spectral sensitivity of the circadian photoreception system may depend on the intensity and duration of light and that there may be changes in spectral sensitivity during the day. While blue light therapy to facilitate circadian adaptation and alertness has been proposed based largely on the effects of melatonin suppression at night, these data may not be transferable to the effects at other times of day (when melatonin is not produced) or for other effects of light. Before blue light is operationalized, three fundamental questions remain to be determined: 1) Does the spectral sensitivity depend on the intensity of light? 2) Do all non-visual effects of light have the same spectral sensitivity? 3) Is the spectral sensitivity of the circadian photoreception system different between the day and the night? Failure to address these questions may result in deployment of suboptimal or less energy-efficient lighting schedules. The proposed project will test whether the spectral sensitivity of the circadian photoreception system is intensity-dependent, response-dependent or time-of-day dependent, by comparing the effects of exposure to monochromatic light during the day and the night, at both high and low intensities, on circadian phase resetting and enhancement of alertness and performance. Experiment 1 aims to compare the melatonin suppression, phase resetting and alerting effects of exposure to dim 460 nm and 555 nm monochromatic light exposure at night, and compare these data to previously collected findings for bright light exposure at night using the same protocol. Data collection for Experiment 1 is complete and analysis is complete for the results for phase shifting, melatonin suppression, subjective alertness and psychomotor performance. Waking EEG data are still under review. Experiment 2 will test the effects of exposure to bright 460 nm and 555 nm monochromatic light during the daytime and is ongoing, with data collection due for completion by the end of 2010. Experiment 3 aims to test the effects of exposure to dim 460 nm and 555 nm monochromatic light during the daytime and will begin in the fall of 2010.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The purpose of the proposed studies was to test three specific hypotheses aimed at evaluating the spectral and temporal sensitivity of light to reset the circadian pacemaker and acutely improve alertness and performance during both day-time and night-time exposures. These studies are designed to provide fundamental information about the photoreceptor systems that detect light for a range of circadian, neuroendocrine, and neurobehavioral responses to light in order that optimal lighting can be designed to alleviate circadian misalignment and acutely improve alertness during space exploration missions. Although the primary goal of the project is to develop lighting countermeasures for the space environment, the funded research project also has important implications for Earth-based applications including light therapy for treatment of circadian rhythm disorders (Advanced and Delayed Sleep Phase Syndromes, Shift-Work Disorder and 'Jet-lag' Disorder) and the use of light as a non-pharmaceutical fatigue countermeasure in many other setting where high alertness and performance is paramount, such as military personnel, first-responders (police, firefighters, paramedics), and other safety-sensitive professions (e.g., pilots, physicians, nurses, truck drivers). Our work has already begun to influence architectural lighting design with the development of methods both by ourselves and others to incorporate the 'non-visual' effects of light into design of 'healthy' buildings, aimed at reinforcing robust circadian entrainment through light-dark cycles, and the potential of light to enhance workplace performance and alertness. Dr. Lockley is also a member of several national and international Committees tasked with developing new lighting standards for the 'non-visual' effects of light which will be informed by the current project and will ultimately be used by industries world-wide to develop lighting and lighting practices that optimize both the visual and non-visual effects of light.
Task Progress:	We have completed Experiment 1 and 18 subjects completed the protocol. Five additional subjects were admitted to the study but withdrew. We had some technical problems with the monochromatic light system required to generate the controlled lighting for the study which delayed recruitment temporarily. We have since taken extra steps to reduce future delays by purchasing a new monochromatic light generation device using other federal funds. Over the past year, we have focused on completing the analysis for Experiment 1 and planning Experiment 2. Data collection for Experiment 1 is complete and analysis is complete for the results for phase shifting, melatonin suppression, subjective alertness and psychomotor performance. We are currently scoring the waking EEG data and anticipate completion of EEG analysis by the end of 2010. Experiment 2 aims to test the effects of exposure to bright 460 nm and 555 nm monochromatic light during the daytime. Over the past year, we have reviewed the protocol and made the decision to reduce the duration of the study from 9 to 7 days. This has led to a small delay in beginning Experiment 2 due to obtaining ethical and institutional approval for the change, rewriting study orders and reprogramming the event schedule software used to co-ordinate events during the study. Experiment 2 is currently actively recruiting subjects and one subject has completed the study to date. We anticipate studying all 16 subjects required to complete Experiment 2 by the end of the year. Experiment 3 aims to test the effects of exposure to dim 460 nm and 555 nm monochromatic light during the daytime. Given the failure to detect major differences between exposure to dim 460 nm and 555 nm monochromatic light at night, we are currently reviewing this aim and, based on the additional data from Experiment 2, we will review and finalize the aims for Experiment 3 by the end of the year, and begin data collection in early 2011.
Bibliography Type:	Description: (Last Updated: 11/14/2025)

Articles in Peer-reviewed Journals	Gooley JJ, Rajaratnam SM, Brainard GC, Kronauer RE, Czeisler CA, Lockley SW. "Spectral responses of the human circadian system depend on the irradiance and duration of exposure to light." Sci Transl Med. 2010 May 12;2(31):31ra33. http://dx.doi.org/10.1126/scitranslmed.3000741 ; PMID: 20463367 , May-2010
Awards	Andersen M, Pechacek CS, Lockley SW. "Taylor Technical Talent Award, Illuminating Engineering Society of North America, July 2009." Jul-2009
Awards	Lockley SW. "Adjunct Associate Professor, Monash University, Australia, July 2009." Jul-2009
Awards	Lockley SW. "Associate Editorial Board, Frontiers in Sleep Disorders, January 2010." Jan-2010
Awards	Lockley SW. "Distinguished Service Award, Sleep Research Society, July 2010." Jul-2010
Awards	Lockley SW. "Harvard Club of Australia Foundation Harvard-Australia Fellowship, October 2009." Oct-2009
Awards	Lockley SW. "Member, International Development Committee, Australasian Sleep Trials Network (ASTN), Australia, January 2010." Jan-2010
Awards	Lockley SW. "Member, Light and Human Health Committee, International Dark-Sky Association, January 2010." Jan-2010
Awards	Lockley SW. "Member, Spaceflight Human Behavior and Performance Working Group (SHBPWG) Subgroup on Circadian Desynchrony, International Space Station Multilateral Medical Operations Panel (MMOP), March 2010." Mar-2010
Awards	Lockley SW. "Member, Technical Committee 169, Non-visual aspects of light on human beings, Comite Europeen de Normalisation (CEN), July 2009." Jul-2009
Awards	Lockley SW. "Preceptor, Training Program in Sleep, Circadian and Respiratory Neurobiology, Division of Sleep Medicine, Harvard Medical School, August 2010." Aug-2010
Awards	Lockley SW. "Research Associate, Woolcock Institute of Medical Research, Australia, January 2010." Jan-2010